Volatility Research on NIFTY-50: Realized Volatility, Persistence, and Implied Volatility Surface Modelling

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Abstract

This paper documents a three-part empirical investigation into equity volatility on the NSE NIFTY-50 index. Part I establishes the statistical properties of realized volatility using rolling-window estimation and OLS persistence regressions. Part II extends this framework to intraday-based estimators — specifically the Garman-Klass variance estimator — and constructs a variance ratio to detect microstructure activity. Part III shifts to the options market, extracting implied volatilities from the NIFTY-50 option chain and fitting a SABR stochastic volatility model to characterize the observed volatility smile. Together, the three parts form a coherent pipeline from raw prices to calibrated volatility surfaces.

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1. Introduction

Volatility occupies a central position in financial economics. It enters option pricing models as the primary unobservable, governs risk-adjusted performance attribution, and drives portfolio construction through its role in covariance estimation. Despite its centrality, volatility is latent — it must be estimated or inferred from observable market data.

This research addresses three interconnected questions using NIFTY-50 data:

1. **Persistence:** Does today's realized volatility predict tomorrow's? How strong is the clustering effect on India's benchmark equity index?
2. **Intraday information:** Does incorporating high-frequency (OHLC) price information improve volatility measurement relative to close-to-close returns?
3. **Implied surface:** What does the NIFTY option chain reveal about market-implied volatility structure, and can the SABR model fit the observed smile?

The analysis is conducted entirely in Python, using publicly available data from `yfinance` and the NSE option chain.

2. Part I — Realized Volatility and Persistence

2.1. Data and Construction

Daily closing prices for the NIFTY-50 index (`^NSEI`) are downloaded from Yahoo Finance for the period January 2022 to March 2026, yielding approximately 1,000 weekly observations after aggregation.

Log returns are computed as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

Realized volatility is estimated using a 5-day rolling standard deviation:

$$\hat{\sigma}_t = \sqrt{\frac{1}{W-1}\sum_{i=0}^{W-1}(r_{t-i}-\bar{r})^2}, \quad W = 5 \tag{2}$$

This is a standard non-parametric estimator that approximates the quadratic variation of the price process over a local window.

2.2. Volatility Persistence

To test for volatility clustering, we estimate the following OLS regression:

$$\hat{\sigma}_t = \alpha + \beta\hat{\sigma}_{t-1} + \varepsilon_t \tag{3}$$

A statistically significant $\beta > 0$ with a high R^2 constitutes evidence of persistence — the empirical hallmark of ARCH-family dynamics. Intuitively, this is the reduced-form analogue of the GARCH(1,1) variance equation.

2.3. Key Findings

- The distribution of realized volatility is right-skewed with a long upper tail, consistent with occasional stress episodes (e.g., rate shock periods, geopolitical events).
- Volatility clustering is visually evident: high-volatility periods tend to cluster together rather than appearing randomly through time.
- The OLS regression confirms a positive, statistically significant slope coefficient $\hat{\beta}$, with moderate R^2, indicating meaningful but imperfect persistence.

3. Part II — Garman-Klass Estimator and Variance Ratio Analysis

3.1. Motivation

Close-to-close (CC) returns discard all intraday price information. The **Garman-Klass (GK)** estimator (1980) exploits the full OHLC price record to produce a lower-variance estimate of the true quadratic variation:

$$\sigma^2_{GK,t} = \frac{1}{2}\left[\ln\frac{H_t}{L_t}\right]^2 - (2\ln 2 - 1)\left[\ln\frac{C_t}{O_t}\right]^2 \tag{4}$$

where H_t, L_t, C_t, O_t denote the daily high, low, close, and open respectively.

3.2. Variance Ratio Construction

To compare the two estimators, we construct a rolling variance ratio:

$$VR_t = \frac{\sum_{i=0}^{9}\sigma^2_{GK,t-i}}{\sum_{i=0}^{9} r^2_{t-i}} \tag{5}$$

This ratio measures the relative contribution of intraday price movement to total daily variance. Values above 1 indicate that intraday range carries more information than overnight close-to-close moves — a signal of heightened intraday activity or microstructure effects.

3.3. Key Findings

- The GK estimator consistently produces smoother volatility series with fewer extreme observations, consistent with its theoretical efficiency advantage.
- The variance ratio fluctuates meaningfully over time, spiking around market stress events — suggesting that intraday price discovery becomes more active during uncertainty.
- Overlaying the variance ratio against the NIFTY price level reveals alignment between ratio spikes and sharp market drawdowns or recoveries, consistent with the GK ratio serving as a regime indicator.

4. Part III — Implied Volatility and SABR Calibration

4.1. Option Chain Data

The NSE NIFTY-50 option chain for the 30 March 2026 expiry is sourced directly from the exchange. The dataset contains, for each strike K: call and put last traded prices (`CE_LTP`, `PE_LTP`), bid-ask spreads, open interest, and — critically — implied volatilities (`CE_IV`, `PE_IV`) as reported by the exchange.

4.2. Implied Volatility Smile

Implied volatility (IV) is extracted for both calls and puts across strikes. Under the Black-Scholes framework, IV would be flat across strikes (the flat surface implied by log-normal prices). In practice, we observe a **volatility smile / skew**:

- Put IV exceeds call IV, especially at lower (out-of-the-money) strikes.
- This put skew reflects market pricing of left-tail crash risk — consistent with elevated geopolitical uncertainty (Iran-US tensions) at the time of observation.

4.3. SABR Stochastic Volatility Model

To fit the observed smile, we calibrate the **SABR model** (Hagan et al., 2002), which specifies:

$$dF_t = \sigma_t F_t^{\beta}\, dW_t^{(1)} \tag{6}$$

$$d\sigma_t = \nu \sigma_t\, dW_t^{(2)} \tag{7}$$

$$\langle dW^{(1)}, dW^{(2)} \rangle = \rho\, dt \tag{8}$$

where F_t is the forward price, σ_t is the stochastic vol, β controls the CEV backbone, ν is the vol-of-vol, and ρ captures the correlation between price and volatility shocks (negative ρ generates the left skew).

The approximate closed-form implied volatility under SABR is:

$$\sigma_{SABR}(K,T) = \frac{\alpha}{(FK)^{(1-\beta)/2}} \cdot \frac{z}{x(z)} \cdot \left[1 + \left(\frac{(1-\beta)^2}{24} \frac{\alpha^2}{(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4(FK)^{(1-\beta)/2}} + \frac{2-3\rho^2}{24}\nu^2 \right) T \right] \tag{9}$$

where:

$$z = \frac{\nu}{\alpha}(FK)^{(1-\beta)/2} \ln \frac{F}{K}$$

$$x(z) = \ln \left(\frac{\sqrt{1 - 2\rho z + z^2} + z - \rho}{1 - \rho} \right)$$

4.4. Calibration

The four SABR parameters $(\alpha, \beta, \rho, \nu)$ are estimated by minimising the sum of squared deviations between model-implied and market-observed IVs:

$$\min_{\alpha,\beta,\rho,\nu} \sum_i \left[\sigma_{SABR}(K_i, T) - \sigma_i^{mkt}\right]^2 \tag{10}$$

using `scipy.optimize.minimize` with β fixed at 0.5 (a common convention for equity indices).

4.5. Key Findings

- The SABR model successfully fits the shape of the NIFTY IV smile, capturing both the ATM level and the downside skew.
- The calibrated $\rho < 0$ confirms the leverage effect: volatility rises when prices fall.
- The calibrated $\nu > 0$ reflects significant vol-of-vol, consistent with the dynamic nature of NIFTY option pricing around macro events.

5. Discussion and Conclusion

This research demonstrates a complete empirical volatility pipeline applied to Indian equity markets:

1. **Realized volatility** on NIFTY-50 exhibits strong persistence and right-skewed distribution, consistent with global equity stylised facts.
2. **Intraday estimators** (Garman-Klass) add information beyond close-to-close returns, with the variance ratio serving as a useful regime indicator.
3. **The NIFTY IV smile** displays significant put skew, reflecting crash-risk pricing that the SABR model captures well through its negative correlation parameter.

Taken together, these findings highlight the richness of NIFTY volatility dynamics and the utility of combining time-series and options-based approaches for a comprehensive view of market risk.

5.1. Extensions

Several natural extensions follow from this work:

- Fitting a full GARCH(1,1) or GJR-GARCH to the realized series and comparing conditional volatility forecasts against the SABR-implied surface.
- Extending the variance ratio analysis across multiple expiries to construct a full term structure of implied volatility.
- Incorporating F&O open interest and options flow as alternative data signals for volatility regime prediction.

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